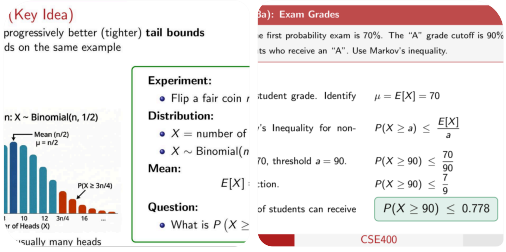
ChatGPT - Lecture scribes



L21_L22.pdf
PDF

I am uploading my lecture slide and some reference images from within the slide itself and I want the Lecture21_22 slide to be used as the only reference material, please help me convert all these slides in the pdf into a lecture scribe which I can use for exam revision. While writing the scribe, go through the material step by step, especially wherever the lecture introduces definitions, assumptions, derivations, proofs, or solved examples. I want the reasoning that is already present in the lecture to be clearly written out in the scribe so that I can understand how each step follows from the previous one during revision. Any definitions or assumptions should be clearly stated before they are applied.If there are mathematical steps, proofs, or worked examples in the slides,present them in a clear step-by-step way, without skipping intermediate steps. Please make sure that all explanations and reasoning are taken only from the uploaded slides, and do not add new examples, alternative explanations, or extra information and the overall tone should be formal and clear, similar to exam-oriented class notes and give all in text

Here is your **exam-oriented lecture scribe** strictly based on the uploaded slides (Lecture 21 & 22: Inequalities and Tail Bounds). All reasoning is expanded step-by-step, without adding external content.

Lecture 21 & 22: Inequalities and Tail Bounds

(CSE400 – Fundamentals of Probability in Computing)

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1. Motivation: Tail Bounds

The lecture begins by introducing the idea of **tail bounds**.

Key Idea:

- We are interested in probabilities of the form:

$$P(X \geq a)$$

  where a is **far from the mean**.
- These probabilities correspond to the **tail of a distribution**.

Interpretation:

- The "tail" represents **rare or extreme events**.
- Example: Getting unusually many heads in coin tosses.

Running Example (Used Throughout the Lecture)

Experiment:

- Flip a fair coin n times.

Random Variable:

- $X =$ number of heads.

Distribution:

$$X \sim \text{Binomial}(n, \tfrac{1}{2})$$

Mean:

$$E[X] = \frac{n}{2}$$

Question:

$$P\left(X \geq \frac{3n}{4}\right)$$

Interpretation:

- This asks: **What is the probability of getting significantly more heads than expected?**
- Since the mean is $n/2$, the value $3n/4$ lies in the **right tail**.

Important Observation:

- We will **not compute exact probabilities**.
- Instead, we will compute **upper bounds**.
- These bounds will become **progressively tighter** using different inequalities.

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2. Markov's Inequality

Definition

Let $X \geq 0$ be a **non-negative random variable**.

Then for any $a > 0$:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Assumptions

- $X \geq 0$
- $E[X]$ exists

Interpretation

- The probability that X is large is bounded by its **mean scaled by** a.
- This is a **very general inequality** (works for any distribution).

Example: Exam Grades

Problem Statement:

- Average grade = 70%
- "A" cutoff = 90%
- Find an **upper bound** on:

$$P(X \geq 90)$$

Step-by-Step Solution

Step 1: Define Random Variable

- Let $X \geq 0$ = student grade

Step 2: Identify Mean

$$\mu = E[X] = 70$$

Step 3: Apply Markov's Inequality

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Step 4: Substitute Values

- $a = 90$

$$P(X \geq 90) \leq \frac{70}{90}$$

Step 5: Simplify

$$P(X \geq 90) \leq \frac{7}{9}$$

$$P(X \geq 90) \leq 0.778$$

Final Result

At most **77.8%** of students can receive an A.

Key Insight:

- This bound is **valid but loose**.
- It does not use any information beyond the mean.

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3. Chebyshev's Inequality

Motivation

- Markov's inequality only uses the **mean**.
- To get tighter bounds, we incorporate **variance**.

Definition

For any random variable X with mean μ and variance σ^2:

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Interpretation

- Measures how far X deviates from its mean.
- Provides a bound on **deviation probability**.

Rewriting for One-Sided Tail

$$P(X \geq \mu + a) \leq \frac{\sigma^2}{a^2}$$

Key Insight

- Uses both:
  - Mean μ
  - Variance σ^2
- Produces a **tighter bound than Markov**.

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4. Chernoff Bound

Motivation

- Even Chebyshev bounds can be loose.
- We want **exponentially decreasing bounds**.

Idea

- Transform the random variable using exponentials.
- Apply Markov's inequality on transformed variable.

Form (General Idea)

For suitable $t > 0$:

$$P(X \geq a) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Key Insight

- This introduces:
  - **Moment Generating Function (MGF)**
- Leads to **very tight bounds**.

For Binomial Distribution

Chernoff bounds give results of the form:

$$P(X \geq (1+\delta)\mu) \leq e^{-c\mu}$$

Interpretation:

- Probability of large deviation **drops exponentially**.

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5. Poisson Tail

Context

- Similar tail bounds apply to **Poisson distributions**.

Key Idea

- Tail probabilities can also be bounded using exponential techniques.

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6. Jensen's Inequality

Definition

If f is a **convex function**, then:

$$f(E[X]) \leq E[f(X)]$$

Convex Function Condition

A function f is convex if:

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2)$$

Interpretation

- Applying function after expectation gives **lower value** than expectation after function.

Importance

- Used in:
  - Deriving Chernoff bounds
  - Bounding expectations

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7. Case Study: System-Level Understanding

Problem

How to predict **worst-case demand on a server?**

Connection to Tail Bounds

- Server load can be modeled as a random variable.
- We want:

$$P(\text{load} \geq \text{capacity})$$

Use of Inequalities

- Markov → rough estimate
- Chebyshev → better estimate
- Chernoff → tight estimate

Conclusion

- Tail bounds help ensure:
  - **Reliability**
  - **System safety**
  - **Performance guarantees**

Final Summary

Inequality	Uses	Strength
Markov	Mean only	Weak
Chebyshev	Mean + Variance	Moderate
Chernoff	Exponential moments	Strong

Key Takeaways

1. Tail bounds estimate probabilities of **rare events**.
2. Markov inequality provides a **basic upper bound**.
3. Chebyshev improves it using **variance**.
4. Chernoff bounds give **exponentially tight bounds**.
5. Jensen's inequality supports advanced derivations.
6. These tools are essential for **analyzing worst-case scenarios**.

If you want next, I can convert this into **perfect LaTeX (exam-ready)** or create a **quick revision cheat sheet**.